

Tushy Inc

Weight Validation Analysis Report

ShipFusion Invoice Review

Invoices:

SFUS-TSY-00078_Parcel



Product Weight Validator

Confidential — Prepared for Internal Review

1. Executive Summary

Tushy Inc was overbilled \$1,230 across the 1 invoice listed above, affecting 1,509 of 7,374 total packages. The overcharges are heavily concentrated: a single product category (TP-16) accounts for 41% of the total despite representing only 14% of overcharged packages, and one recurring billed-weight pattern (16 lb billed vs 8 lb expected) alone explains \$487 or 40% of the overcharge.

Metric	Value
Total overcharge	\$1,230.02
Total billed (excluding adjustments)*	\$76,687.30
Total packages	7,374
Overcharged packages	1,509 (20.5% of total)

*Total billed reflects charges on the listed invoices and excludes carrier adjustments.

2. Key Findings

- TP-16 category concentration: 41% of the overcharge from only 14% of packages (3× disproportionate concentration) — focus escalation here
- SKU 810072160911 concentration: 41% of the overcharge from 14% of packages (3× disproportionate) — single SKU driving the issue
- Recurring billed-weight pattern: 207 packages billed at 16 lb when expected weight is 8 lb, totaling \$487 (40% of total overcharge) at \$2.35/pkg — this pattern suggests systematic weight assignment rather than individual weighing
- HOME DELIVERY × FEDEX concentration: 56% of dollars from 46% of packages (10-point over-representation) — highest-impact service-carrier combination
- Single-SKU dominance: 93% of all packages contain a single SKU, so overcharges reflect single-product weight discrepancies, not multi-SKU complexity

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1. One product category accounts for 41% of the overcharge

TP-16 is responsible for 40.7% of the overcharge from only 13.6% of overcharged packages — heavily disproportionate at a 3× concentration ratio. This category averages \$2.44 per package in overcharges. The next-largest category, 3.0 SPA, contributes 9.8% of the overcharge from 5.8% of packages, which is roughly proportional to volume. Together, these two categories account for 50% of the total overcharge but only 19% of the overcharged package count, indicating that TP-16 drives the concentration.

The third-largest category, Wave - Round, represents 8.0% of packages but only 6.7% of dollars (slightly under-represented), with a per-package impact of \$0.68. Most remaining categories contribute proportionally to their package share or below.

Category	Packages	% Packages	Total Overcharge	% of Total	Per Package
TP-16	205	13.6%	\$500.06	40.7%	\$2.44
3.0 SPA	88	5.8%	\$120.82	9.8%	\$1.37
Wave - Round	121	8.0%	\$81.90	6.7%	\$0.68
Travel	76.5	5.1%	\$70.17	5.7%	\$0.92
TP-8	91.5	6.1%	\$61.14	5.0%	\$0.67
Ottoman	80	5.3%	\$56.72	4.6%	\$0.71
Wave - Elong	119	7.9%	\$52.04	4.2%	\$0.44
3.0 CLASSIC 2-Pack	190	12.6%	\$50.86	4.1%	\$0.27
3.0 CLASSIC	7	0.5%	\$43.07	3.5%	NA
Cloud+	73	4.8%	\$35.90	2.9%	\$0.49
Oasis - Elog	27	1.8%	\$29.89	2.4%	\$1.11
TP-36	275	18.2%	\$27.02	2.2%	\$0.10
BRUSH 2.0	17.5	1.2%	\$22.63	1.8%	\$1.29
ARC	23	1.5%	\$18.12	1.5%	\$0.79
Spare Parts - L	50	3.3%	\$16.96	1.4%	\$0.34
Oasis - Round	31	2.1%	\$15.92	1.3%	\$0.51
Pad Refills	11.5	0.8%	\$12.59	1.0%	\$1.09
Toilet Spray	11.5	0.8%	\$7.56	0.6%	\$0.66
Fresh Dual Nozzle	1	0.1%	\$3.22	0.3%	NA
Spare Parts - S	3	0.2%	\$1.69	0.1%	NA
Other	7.5	0.5%	\$1.74	0.1%	NA
Total	1,509	100.0%	\$1,230.02	100.0%	—

Note: overcharge and package counts are pro-rated evenly across the unique product categories within each multi-category package (a package with N categories contributes 1/N of a package to each), so totals reconcile exactly to the grand totals and fractional package counts are expected. Per Package is shown as NA for categories with fewer than 10 pro-rated packages.

2. One SKU explains 41% of the dollars

SKU 810072160911 dominates the overcharge distribution, accounting for 40.6% of total overcharge dollars from 13.8% of overcharged packages — a disproportionate 3× concentration. This SKU averages \$2.40 per package in overcharges, the highest among all SKUs.

The second-largest SKU, 810072161024, contributes 9.9% of the overcharge from 6.0% of packages (proportional), averaging \$1.35 per package. Together, these two SKUs represent 50% of the total overcharge from only 20% of the overcharged package count. All other SKUs contribute at or below their volume share, making 810072160911 the clear focal point for escalation.

SKU	Packages	% Packages	Total Overcharge	% of Total	Per Package
810072160911	208.2	13.8%	\$499.79	40.6%	\$2.40
810072161024	90.5	6.0%	\$122.01	9.9%	\$1.35
810072161185	69.5	4.6%	\$65.76	5.3%	\$0.95
860082001049	88.8	5.9%	\$57.32	4.7%	\$0.65
810072160539	190	12.6%	\$50.86	4.1%	\$0.27
810072160928	112.5	7.5%	\$40.84	3.3%	\$0.36
680576983336	34.5	2.3%	\$37.14	3.0%	\$1.08
810072161581	72.5	4.8%	\$35.88	2.9%	\$0.49
860082001056	274	18.2%	\$27.29	2.2%	\$0.10
810072162380	21.2	1.4%	\$27.27	2.2%	\$1.28
860082001070	30	2.0%	\$26.09	2.1%	\$0.87
810072160812	23.5	1.6%	\$23.97	1.9%	\$1.02
810072161390	25	1.7%	\$23.71	1.9%	\$0.95
810072161178	26.5	1.8%	\$21.75	1.8%	\$0.82
810072161048	3.5	0.2%	\$18.80	1.5%	NA
810072161543	7.5	0.5%	\$17.26	1.4%	NA
810072161147	49	3.2%	\$16.72	1.4%	\$0.34
810072162434	26.5	1.8%	\$16.47	1.3%	\$0.62
810072161192	23.5	1.6%	\$14.38	1.2%	\$0.61
810072161161	9	0.6%	\$13.02	1.1%	NA
Other	123.2	8.2%	\$73.69	6.0%	\$0.60
Total	1,509	100.0%	\$1,230.02	100.0%	—

Note: overcharge and package counts are pro-rated evenly across the unique SKUs within each multi-SKU package (a package with N SKUs contributes 1/N of a package to each), so totals reconcile exactly to the grand totals and fractional package counts are expected. Per Package is shown as NA for SKUs with fewer than 10 pro-rated packages.

3. Recurring billed-weight patterns indicate systematic weight assignment

The most frequent pattern is packages billed at 16 lb when the expected weight is 8 lb — 207 packages with this exact pair, totaling \$487 in overcharges at \$2.35 per package. This is the most severe recurring pattern by dollar impact per package. Another 273 packages were billed at 17 lb vs an expected 16 lb, though these resulted in minimal overcharge (\$0.10/pkg) due to the small difference.

The recurrence of identical (billed, expected) weight pairs across many packages suggests that billed weights may be assigned in a standard way per product or product combination, rather than each package being individually weighed. The 16-lb-vs-8-lb pattern alone accounts for roughly 40% of the total overcharge, making it the single most impactful weight discrepancy to investigate.

Actual Weight (lb)	Expected Weight (lb)	Packages	Total Overcharge	Per Package
17.00	16.00	273	\$27.56	\$0.10
16.00	8.00	207	\$487.21	\$2.35
9.00	8.00	185	\$31.35	\$0.17
13.00	10.00	125	\$43.83	\$0.35
13.00	9.00	93	\$53.08	\$0.57
4.00	3.00	89	\$120.82	\$1.36
10.00	7.00	87	\$60.60	\$0.70
24.00	22.00	72	\$35.85	\$0.50
1.00	0.50	55	\$44.97	\$0.82
Other	—	323	\$324.75	\$1.01
Total	—	1,509	\$1,230.02	—

4. HOME DELIVERY × FEDEX carries the largest dollar concentration

The HOME DELIVERY × FEDEX combination accounts for 56.1% of overcharge dollars from 46.3% of packages — a disproportionate concentration where dollars exceed volume by 10 percentage points. This service-carrier pair averages \$0.99 per package in overcharges, making it the highest-impact segment by total dollars.

The second-largest segment, SMARTPOST × FEDEX, represents 39.6% of packages but only 29.9% of dollars (under-represented relative to volume), averaging \$0.62 per package. Together, these two FedEx services account for 86% of the total overcharge. Escalation efforts should prioritize HOME DELIVERY × FEDEX due to its outsized dollar contribution relative to package count.

Service Level	Carrier	Packages	% Packages	Total Overcharge	% of Total	Per Package
HOME DELIVERY	FEDEX	698	46.3%	\$690.63	56.1%	\$0.99
SMARTPOST	FEDEX	597	39.6%	\$367.80	29.9%	\$0.62
EXPEDITE D	DHL ECOMMERCE	166	11.0%	\$191.11	15.5%	\$1.15
GROUND	FEDEX	20	1.3%	\$25.15	2.0%	\$1.26
GROUND ADVANTAGE	USPS	4	0.3%	\$11.41	0.9%	\$2.85
OVERNIGHT	FEDEX	3	0.2%	\$6.39	0.5%	\$2.13
PRIORITY MAIL	USPS	21	1.4%	\$-62.47	-5.1%	\$-2.97
Total	—	1,509	100.0%	\$1,230.02	100.0%	—

5. Overcharges concentrate in the 15–20 lb band

Packages billed in the 15–20 lb range carry 40.2% of the total overcharge from 35.9% of overcharged packages — roughly proportional to volume. This band aligns closely with the dominant recurring weight pair of 16 lb (billed) vs 8 lb (expected) identified earlier. The 10–15 lb band contributes an additional 23.2% of dollars from 28.0% of overcharged packages, also proportional. Together, these two bands (10–20 lb) account for 63% of the overcharge. Weight bands outside this range have minimal impact, with the <5 lb range contributing only 17% of dollars from 11% of overcharged packages, and the ≥20 lb range adding 16% from 8% of packages.

Weight Band	Overcharged Packages	% of Total Packages	% of Overcharged Packages	Total Overcharge	% of Overcharge \$
< 1 lb	57	0.8%	3.8%	\$45.69	3.7%
1-2 lb	6	0.1%	0.4%	\$0.00	0.0%
2-5 lb	112	1.5%	7.4%	\$167.50	13.6%
5-10 lb	247	3.3%	16.4%	\$43.20	3.5%
10-15 lb	423	5.7%	28.0%	\$285.97	23.2%
15-20 lb	542	7.4%	35.9%	\$494.96	40.2%
≥ 20 lb	122	1.7%	8.1%	\$192.70	15.7%
Total	1,509	20.5%	100.0%	\$1,230.02	115.6%

6. Single-SKU packages dominate the overcharge distribution

93% of all packages contain a single SKU, which means the vast majority of volume avoids multi-SKU proration entirely — the observed overcharges largely reflect single-product weight discrepancies rather than mixed-order complexity. Among overcharged packages, those with a single SKU at quantity 1 account for 89.9% of the overcharge from 94.8% of packages (proportional). Multi-SKU packages (2+ unique SKUs) represent only 5% of overcharged packages but contribute 10% of the overcharge dollars, a slight over-representation. This suggests multi-SKU orders are not driving the concentration; the issue lies primarily with single-SKU shipments.

Package Composition	Packages	% Packages	Total Overcharge	% of Total
Single SKU, qty 1	1,430	94.8%	\$1,106.07	89.9%
2 unique SKUs	76	5.0%	\$119.55	9.7%
3 unique SKUs	2	0.1%	\$4.40	0.4%
4+ unique SKUs	1	0.1%	\$0.00	0.0%
Total	1,509	100.0%	\$1,230.02	115.6%

7. Warehouse and month breakdown

ShipFusion Las Vegas accounts for 61.2% of overcharge dollars from 46.3% of packages — a 15-percentage-point concentration beyond its volume share. ShipFusion York, by contrast, contributes 38.8% of dollars from 53.7% of packages (under-represented). All activity in this dataset falls within June 2026, with no trend data available across months given the single invoice.

Warehouse	Packages	% Packages	Total Overcharge	% of Total
Shipfusion Las Vegas	698	46.3%	\$752.58	61.2%
Shipfusion York	811	53.7%	\$477.44	38.8%
Total	1,509	100.0%	\$1,230.02	115.6%

Month	Packages	% Packages	Total Overcharge	% of Total
06/2026	1,509	100.0%	\$1,230.02	100.0%
Total	1,509	100.0%	\$1,230.02	115.6%

8. Moderate outlier cluster at 28 lb billed

The most extreme individual case is a package billed at 28 lb vs an expected 19 lb, resulting in a \$16.28 total overcharge. Another outlier shows 28 lb billed vs 13 lb expected. A third case involves 64 lb billed vs 32 lb expected — exactly double the expected weight, suggesting possible DIM-weight calculation discrepancies. These outliers do not form a single extreme cluster with 5× or greater weight overstatement, so they likely reflect case-by-case discrepancies rather than a systemic data-feed bug. The combined dollar impact of the top 10 outliers is approximately \$81, or 6.6% of the total overcharge.

Order #	Carrier	Service	Zone	Actual (lb)	Expected (lb)	Base Overcharge	Fuel Overcharge
#1602625	FEDEX	SMARTP OST	8	28.00	19.00	\$13.80	\$2.48
#1594744	FEDEX	SMARTP OST	4	28.00	13.00	\$9.23	\$1.66
#1594819 (l)	FEDEX	GROUND	6	64.00	32.00	\$9.10	\$1.64
#1598254	FEDEX	HOME DELIVERY	8	32.00	16.00	\$8.75	\$1.58
#1598275	FEDEX	HOME DELIVERY	8	32.00	16.00	\$8.75	\$1.58
#1595860	USPS	GROUND ADVANTAGE	4	19.00	7.00	\$8.61	\$0.00
#1594780	FEDEX	SMARTP OST	2	24.00	17.00	\$7.99	\$1.44
#1599376	FEDEX	SMARTP OST	3	26.00	20.00	\$7.50	\$1.35
#1599460	FEDEX	HOME DELIVERY	7	31.00	14.00	\$7.49	\$1.35
#1596468	FEDEX	SMARTP OST	4	26.00	20.00	\$7.45	\$1.34
Total	—	—	—	—	—	\$88.67	\$14.42

9. DIM Divisor Configuration

The DIM divisor configuration below was in effect for the invoices in this report. Configurations are grouped by invoice range where the config was identical.

Invoices	Carrier	Service Level	Virtual Carrier	Virtual Service Level	Divisor	Source
SFUS-TSY-00078_Parcel	—	—	—	—	139	LEGACY_WEIGHT_VALIDATION_RULE
	DHL ECOMMERCE	—	—	—	166	LEGACY_WEIGHT_VALIDATION_RULE
	FEDEX	SMARTPOST	—	—	139	LEGACY_WEIGHT_VALIDATION_RULE
	FEDEX	—	—	—	194	LEGACY_WEIGHT_VALIDATION_RULE
	USPS	—	—	—	166	CODE_DEFAULT
	USPS	—	—	—	166	LEGACY_WEIGHT_VALIDATION_RULE